Day 4: Other Protocols in Action Theory (30 min)

1.Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Aim

To install FileZilla Server on a Windows machine, create an FTP user account, and configure a shared folder for file transfer.

Theory

FileZilla Server is an FTP server application that allows users to transfer files between computers over a network. It supports user authentication and shared-folder permissions to control access to files.

Procedure

Step 1: Download and Install FileZilla Server

1. Open the official website: https://filezilla-project.org/
2. Download FileZilla Server for Windows.
3. Run the installer and follow the installation instructions.
4. Complete the installation and open the FileZilla Server interface.

Step 2: Create a Shared Folder

1. Create a folder on your computer, for example, C:\FTPShare.
2. Add a sample text file to the folder.
3. This folder will be shared with authorized FTP users.

Step 3: Create an FTP User Account

1. Open the FileZilla Server administration interface.
2. Navigate to the Users section.
3. Click Add to create a new user.
4. Enter a username, such as ftpuser.
5. Enable password authentication and set a password.

Step 4: Configure Shared Folder Access

1. Open the user's shared folders or mount-points settings.
2. Add the folder C:\FTPShare.
3. Assign it as the accessible FTP directory.
4. Enable Read permission to download files. Enable Write only if uploading files is required.
5. Apply or save the configuration.

Step 5: Test the FTP Server

1. Find your computer's IPv4 address by opening Command Prompt and running:

ipconfig

2. Open FileZilla Client or another FTP client.
3. Enter the following connection details:

   | Field | Example | |---|---|

   | Host | 127.0.0.1 (same computer) |

   | Port | 21 (if configured for FTP) |

   | Username | ftpuser |

   | Password | Your configured password |

4. Click Quickconnect and check whether the shared folder and its files are accessible.

2.Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Aim

To connect to the FTP server using FileZilla Client and upload an image file from the computer to the shared folder.

Procedure

Step 1: Open FileZilla Client

1. Download and install FileZilla Client from https://filezilla-project.org/ if it is not already installed.
2. Open FileZilla Client on your Windows computer.

Step 2: Connect to the FTP Server

Enter the following details in the Quickconnect bar:

Field	**Value**
Host	127.0.0.1
Username	ftpuser
Password	Your configured password
Port	21

Click Quickconnect to connect to the FTP server running on the same computer.

Step 3: Select an Image File

1. In the left panel (Local site), navigate to the folder containing an image on your computer.
2. Select any image file, for example, photo.jpg.

Step 4: Upload the Image

1. In the right panel (Remote site), open the FTP shared folder.
2. Drag photo.jpg from the left panel to the right panel.
3. Wait for the transfer to complete.
4. Check the transfer queue at the bottom to confirm that the upload was successful.

Observation

- FileZilla Client successfully connected to the FTP server using the configured credentials.
- The selected image file was transferred from the local computer to the remote FTP shared folder.
- The uploaded image appeared in the Remote site panel.

3.Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

Aim

To download an uploaded image file from the FTP server to a different folder using FileZilla Client and verify that the downloaded file size matches the original file.

Procedure

Step 1: Open FileZilla Client

1. Launch FileZilla Client on your Windows computer.
2. Connect to the FTP server using your configured credentials.

Field	**Value**
Host	127.0.0.1
Username	ftpuser
Password	Your configured password
Port	21

Step 2: Locate the Uploaded File

1. In the right panel (Remote site), open the FTP shared folder.
2. Locate the image file uploaded in Task 2, for example, photo.jpg.

Step 3: Select a Different Download Folder

1. In the left panel (Local site), navigate to a different folder on your computer.
2. For example, create a folder named C:\FTPDownloads.
3. Open this folder in the Local site panel.

Step 4: Download the File

1. Right-click photo.jpg in the Remote site panel.
2. Select Download.
3. Wait until the transfer completes.
4. Verify that photo.jpg appears in C:\FTPDownloads.

Step 5: Verify the File Size

1. Open File Explorer.
2. Right-click the original image and select Properties.
3. Note its size in bytes.
4. Repeat the same process for the downloaded image.
5. Compare the two file sizes.

Alternatively, use Command Prompt:

dir "C:\FTPDownloads\photo.jpg"

Observation

Parameter	**Original File**	**Downloaded File**
File name	photo.jpg	photo.jpg
Location	Original image folder	C:\FTPDownloads
File size	Record actual size	Record actual size
Status	Available	Downloaded

Verification: If both files show the same size in bytes, the file-size verification is successful.

**4.Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Hint: Think about how network admins monitor device health, bandwidth, or uptime.**

Aim

To identify three real-world scenarios where SNMP (Simple Network Management Protocol) is used to monitor and manage network devices, bandwidth, and uptime.

Theory

SNMP (Simple Network Management Protocol) is a network protocol used by network administrators to monitor and manage devices such as routers, switches, servers, printers, and access points. It helps identify network problems and maintain reliable connectivity.

Three Real-World Scenarios

1. Internet Service Providers (ISPs)

ISPs use SNMP to monitor routers, switches, and other networking equipment. They can check bandwidth usage, interface errors, device temperature, and network uptime. If a router stops responding or a connection becomes overloaded, administrators can investigate and resolve the issue.

Example: An ISP monitoring the network that provides internet access for watching YouTube or using Instagram.

2. Campus Wi-Fi Management

Colleges and universities use SNMP to monitor Wi-Fi access points, switches, and network controllers across classrooms, laboratories, and libraries. It helps administrators identify overloaded access points, failed devices, and unusual bandwidth consumption.

Example: When many students connect to campus Wi-Fi simultaneously, SNMP monitoring can help IT staff identify access points experiencing high traffic.

3. Large Websites and Online Services

Large online services, such as Amazon or Flipkart, operate extensive network infrastructure. Their infrastructure teams can use SNMP to monitor supported network switches, routers, and other devices for traffic levels, interface errors, and device availability. This helps them detect network issues that could affect website performance.

Example: During a major online sale, network administrators can monitor traffic on network equipment and identify overloaded links that may affect access to the website.

5.Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Aim

To compare FTP (File Transfer Protocol) and HTTP (Hypertext Transfer Protocol) based on port number, security, data transfer method, use cases, and authentication.

Theory

FTP is used to transfer files between computers and servers, whereas HTTP is used to transfer web pages and other web resources between clients and web servers.

Comparison Table

Sr. No.	Comparison Point	FTP	HTTP
1	Full Form	File Transfer Protocol	Hypertext Transfer Protocol
2	Port Number	Port 21 for control; active FTP commonly uses port 20 for data	Port 80 by default
3	Security	Standard FTP does not encrypt data or credentials	Standard HTTP does not encrypt data
4	Secure Version	FTPS or SFTP (different protocols)	HTTPS, which uses TLS encryption
5	Data Transfer Method	Transfers files using separate control and data connections	Transfers requests and responses between clients and servers
6	Use Cases	Uploading, downloading, and managing files on a server	Accessing websites, web pages, and web APIs
7	Authentication	Can use a username and password; anonymous access may also be allowed	Often uses sessions, tokens, cookies, or other authentication methods; public pages may require none
8	Common Applications	FileZilla Client and FileZilla Server	Web browsers such as Chrome and Firefox

9	Typical Example	Uploading an image to an FTP shared folder	Opening https://www.flipkart.com in a browser